

1. Provide very brief answers to the following questions:

1×6=6

- (a) What is the sum of the weights of a smoothing filter?
- (b) What is the effect of smoothing filter size on blurring?
- (d) What does the quantization process of image digitization do?
- (e) Which mathematical operation is used in smoothing filters?
- (f) Graphically represent the image thresholding operation with a single threshold.

2. Differentiate between:

2×2=4

- (a) Gray scale image and coloured image.
- (b) Point processing technique and filtering technique.

3. (a) What is a filter. Provide examples.

2

(b) How do you filter an image?

2

(c) Provide 3×3 filter masks for the following smoothing filters:

2

- i) Box filter
- ii) Binomial filter

(d) Perform the following filtering operations on the image given below.

5	1	4	6
4	7	2	3
1	2	3	5

i) Apply the binomial filter to determine smoothed value of the pixel at location (1, 1).

2

ii) Apply the 3×3 median filter to determine smoothed value of the pixel at location (1, 2).

2

4. (a) Calculate the 1st order and the 2nd order derivatives at the pixel $f(x, y)$ of an image.

1+2=3

(b) Find first order derivative of the image:

3	7	4	4	6
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2

(c) A sharpened image is not directly obtained by filtering an image with a Laplacian filter. How do you get the sharpened image?

2

(d) Sharpen the pixel $f(1,1)$ of the image given in question no. 3(d) by applying a Laplacian filter.

Present the filter used.

3

5. (a) Provide sample plot of histogram of a dark image and a bright image.

2

(b) What is histogram equalization? What is the need for histogram equalization?

3

(c) Create a 4×4 image with pixel values in the range 0-7.

2

(d) Perform histogram equalization of the image you have created in (b) above.

3